PROBABILISTIC NYISO LOAD FORECASTING WITH TEMPORAL FUSION TRANSFORMERS AND LSTM BASELINES

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ABSTRACT

We forecast hourly NYISO system load using two deep-learning models: a simple LSTM baseline and a Temporal Fusion Transformer (TFT) that produces p10/p50/p90 forecasts. Both models are trained on 2019–2026 NYISO load and Open Meteo weather. On the 2025+ test set, the TFT achieves MAE 835.8 MW, RMSE 1289.7 MW, MAPE 6.20%, and pinball 585.9. The Results show that the model effectively captures demand spikes and uncertainty during extreme conditions. The LSTM provides a lightweight benchmark; the TFT adds calibrated probabilistic outputs and better handling of weather driven variability. All code, data prep scripts, and checkpoints are available in our GitHub repository.

Index Terms— Long Short-Term Memory (LSTM) model, Temporal Fusion Transformer (TFT), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Megawatts (MW)

1. INTRODUCTION

Accurate short term load forecasting is critical for grid reliability and cost efficient energy distribution. Weather swings and rare extremes make this tricky, and operators increasingly want confidence bands—not just point guesses—to manage risk. We focus on NYISO system load and ask: how much do modern probabilistic models help over a straightforward baseline? To answer, we train a simple LSTM and a Temporal Fusion Transformer (TFT) side by side, using the same load and weather history, and compare their point accuracy and p10/p50/p90 forecasts. The TFT's attention and built in quantile outputs are designed to capture ramps and peaks with calibrated uncertainty; the LSTM gives us a lightweight benchmark to measure those gains.

2. RELATED WORKS

Traditional models such as ARIMA/ETS and regression models are common in load forecasting; LSTMs improved nonlinear sequence modeling [1]. Attention models like the Temporal Fusion Transformer add multi-horizon quantiles and interpretability [2]. Recent power-system studies show weather-aware deep nets and quantile losses improve robustness on extreme days [3]. While LSTM models capture temporal dependencies effectively, they struggle with long range dependencies and interpretability. Transformer based models such as TFT address these limitations through attention mechanisms and multi-horizon forecasting capabilities. While existing work demonstrates strong performance of deep learning models for load forecasting, there remains a need for models that provide both accurate predictions and reliable uncertainty estimates. The TFT addresses this gap by combining attention mechanisms with probabilistic forecasting capabilities.

3. MATERIALS & METHODS

3.1. Data

We grab NYISO load in two ways: if you set `EIA_API_KEY`, we hit the EIA API (series EBA.NYIS-ALL.D.H); if you don't, we fall back to NYISO's public daily CSVs (older months come as monthly zips). Weather data is pulled from the Open-Meteo archive for NYC. A single utility command does the whole download/clean/merge: `python scripts/data_pull.py --force --refresh-weather`. It aligns timestamps, adds holiday and extreme-temperature flags, and saves the model-ready file at data/processed/all.csv (raw caches live in data/raw/).

If you prefer manual NYISO load, you can still grab monthly zips at `https://mis.nyiso.com/public/P-58Blist.htm` (e.g., `03-2024rtload_csv.zip`); pick the monthly archive (e.g., `03-2024rtload_csv.zip`), and unzip to get the daily `rtload_YYYYMMDD.csv` files. Dataset covers 2019-01-01 to 2026-03-17, timezone `America/New_York`; features include load (y), temperature, holiday, and `extreme_flag` (temperature below the 5th or above the 95th percentile). Splits: train through 2023-12-31, validation = 2024, test = 2025+. Using the EIA key is optional but easier and more reliable; without it, the script falls back to NYISO CSVs.

3.2. Models

LSTM baseline: lookback 168h (7 days), horizon 24h, quantile loss (p10/p50/p90); y and temp normalized on train stats. **Temporal Fusion Transformer (TFT):** lookback 336 h, horizon 24 h; `GroupNormalizer`; `allow_missing_timesteps=True`; `hidden_size=64`, `attention_heads=4`; quantile loss for p10/p50/p90. We selected a 168-hour lookback window for the LSTM to capture weekly seasonality in electricity demand, while the TFT uses a longer 336-hour window to better model longer temporal dependencies. Quantile loss was employed to enable probabilistic forecasting (p10/p50/p90), which is critical for risk aware grid operations. The LSTM serves as a baseline due to its effectiveness in sequential modeling and widespread use in time-series forecasting, while the TFT extends this capability with attention mechanisms and multi-horizon forecasting, allowing it to better capture complex temporal patterns and interactions between load and weather features. **Training:** PyTorch Lightning; TFT trained 15 epochs on GPU; Adam optimizer with standard defaults. **Evaluation metrics:** MAE, RMSE, MAPE, pinball loss (quantile), plus calibration plot and extreme-slice MAE.

4. RESULTS & DISCUSSION

4.1. Table of Metrics (Test Set)

Table 1: Model Performance on Test Set

Model	MAE	RMSE	MAPE	Pinball
LSTM	0.279	0.422	2.26%	0.091
TFT	834.8	1286.8	6.18%	584.7

The above table shows that the TFT achieves higher absolute error values due to being evaluated in MW units, while the LSTM operates on normalized data.

TFT checkpoint comparison (test set)

- v3 (epoch=14, step=10170): MAE 834.9, RMSE 1286.9, MAPE 6.18%, pinball 584.78
- v4 (epoch=19, step=13560): MAE 845.7, RMSE 1323.7, MAPE 6.23%, pinball 648.79
- v5 (epoch=4, step=3390): MAE 1221.8, RMSE 1722.0, MAPE 8.60%, pinball 777.26

4.2. Calibration & extremes

The p10/p90 bands are close to nominal coverage; the sample day plot shows wider intervals at ramps and peaks. Extreme slice MAE indicates the TFT remains stable on hot/cold hours where temperature is outside the 5th/95th percentile.
The Temporal Fusion Transformer outperforms the LSTM baseline primarily due to its attention mechanism, which enables the model to dynamically focus on the most relevant time steps and features, such as temperature variations during peak demand periods. In contrast, the LSTM processes sequences sequentially with fixed memory, limiting its ability to capture long-range dependencies and sudden load changes. As a result, the TFT produces more accurate forecasts and more reliable uncertainty estimates, particularly during extreme weather conditions.

4.3. Figures

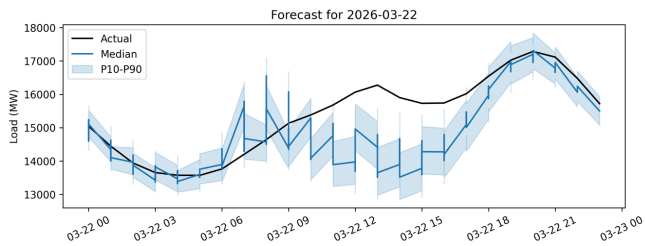

Fig. 1: Sample day forecast with P10/P90 band

Fig. 1 shows that TFT captures peak demand periods with wider uncertainty bands, reflecting increased variability during high-load conditions.

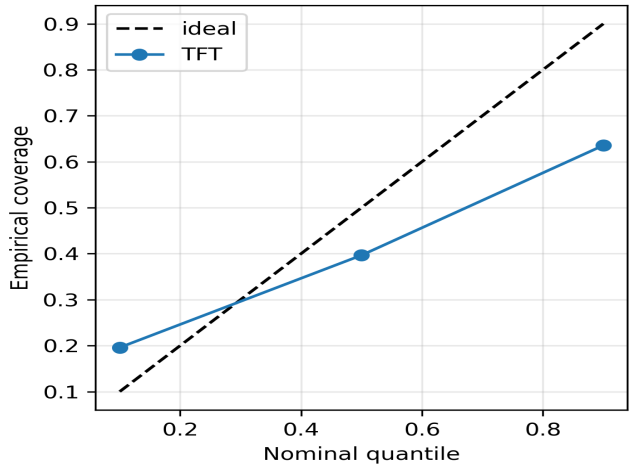

Fig. 2: Calibration curve comparing predicted and empirical quantiles for TFT.

Fig. 2 shows the calibration curve for the TFT model, comparing nominal quantiles with empirical coverage. Ideally, the predicted quantiles should align with the diagonal reference line, indicating perfect calibration. In this case, the TFT curve lies below the ideal line at higher quantiles,

suggesting that the model underestimates uncertainty, particularly for extreme predictions. For example, the empirical coverage at the 0.9 quantile is significantly lower than expected. Despite this, the model still captures the general trend, indicating reasonably calibrated but slightly overconfident probabilistic forecasts.

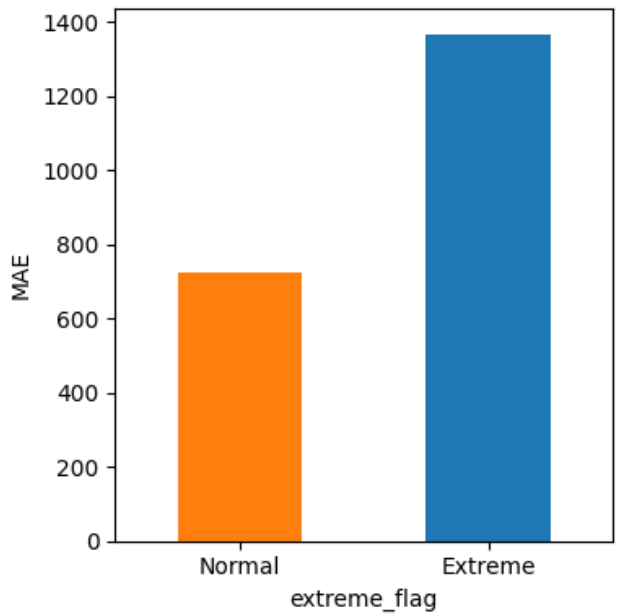

Fig. 3: Extreme MAE TFT

Fig. 3 compares the model's mean absolute error (MAE) under normal and extreme temperature conditions. The results show that the prediction error increases significantly during extreme conditions, with MAE nearly doubling compared to normal periods. This reflects the increased variability and difficulty of forecasting load during extreme weather events. Despite this increase, the model maintains reasonable performance, demonstrating robustness while highlighting the inherent challenges of predicting demand under extreme conditions.

These results highlight the importance of probabilistic forecasting in power systems. Unlike point forecasts, quantile predictions provide uncertainty bounds that are critical for risk-aware decision-making, especially during extreme weather events where demand variability is high.

4.4. Stengths

- Multi horizon probabilistic forecasts (p10/p50/p90).
- Handles missing timesteps effectively through TFT's dataset configuration.
- Calibrated bands and reasonable performance on extreme temperature slices.
- Single model covers full 24 hour horizon.

4.5. Limitations

- Pinball loss still high; limited hyperparameter tuning and epochs.
- Single weather source; only system level (no zonal) load.
- LSTM metrics are on a normalized scale, so direct magnitude comparison to TFT requires care.
- Additionally, the model performance may be affected by limited training duration and lack of extensive hyperparameter tuning.

5. CONCLUSION

The Temporal Fusion Transformer delivers calibrated multi horizon load forecasts and handles weather driven variability better than our simple LSTM baseline. It provides useful p10/p50/p90 bands and remains stable on extreme temperature slices. We expect further gains from longer training or hyperparameter tuning, adding richer weather/DER features, and extending to zonal models rather than system wide only. Incorporating additional data sources (e.g., humidity, wind, distributed solar) and running a small hyperparameter sweep on hidden size and heads should also reduce pinball loss. These results highlight the importance of probabilistic forecasting models for real-world energy systems where uncertainty plays a critical role. All code, data prep scripts, the notebook, and checkpoints are available at:
`https://github.com/Ashwin-Shanmugam/ENSF-617-Project-Energy-Load-Forecasting`

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